Problem 4

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Problem 1 *Determine the following information about the given function and then graph the function:*

$$f(x) = 3\sin\left(\frac{x}{2}\right), [-2\pi, 2\pi]$$

Domain
x, y-intercepts
Symmetry
Asympotes
Intervals of increasing/decreasing
Maxima/Minima
Inflection points
Intervals of concavity
Sketch a graph of f

Explanation. (a) Domain: This was given as $[-2\pi, 2\pi]$

(b) x, y-intercepts:
x-interpects occur when $f(x) = 3\sin\left(\frac{x}{2}\right) = 0 \implies \frac{x}{2} = 0 + k\pi$ where k is an integer, $-1 \leq k \leq 1$. On the domain this is $x = -2\pi, 0, 2\pi$.
y-intercepts: $f(0) = 3\sin\left(\frac{0}{2}\right) = 0$ The y-intercept is the point $(0, 0)$

(c) Symmetry:
$f(-x) = 3\sin\left(\frac{-x}{2}\right) = -3\sin\left(\frac{x}{2}\right) \neq f(x)$ so f is not even $-f(x) = -3\sin\left(\frac{x}{2}\right) = f(-x)$ so f is odd

(d) Intervals of increasing/decreasing:

$$f'(x) = 3\cos\left(\frac{x}{2}\right) \cdot \frac{1}{2} = \frac{3}{2}\cos\left(\frac{x}{2}\right)$$

Since f' is defined everywhere on $(-2\pi, 2\pi)$ to find the critical points we just solve the equation $f'(x) = 0$ where $-2\pi < x < 2\pi$:

$$\begin{aligned} f'(x) = 0 &\iff \frac{3}{2}\cos\left(\frac{x}{2}\right) = 0 \\ &\iff \cos\left(\frac{x}{2}\right) = 0 \\ &\iff -2\pi < x < 2\pi \text{ and } x/2 = \pi/2 + n\pi \text{ with } n \text{ an integer} \\ &\iff -2\pi < x < 2\pi \text{ and } x = \pi + n2\pi \text{ with } n \text{ an integer} \\ &\iff x = -\pi \text{ and } x = \pi. \end{aligned}$$

So the only critical points are $x = -\pi$ and $x = \pi$.

If $-2\pi < x < -\pi$, it follows that $-\pi < x/2 < -\pi/2$ so $x/2$ lies in the third quadrant and $\cos(x/2) < 0$.
If $-\pi < x < \pi$, it follows that $-\pi/2 < x/2 < \pi/2$ so $x/2$ lies in the fourth and first quadrants and $\cos(x/2) > 0$.
If $\pi < x < 2\pi$, it follows that $\pi/2 < x/2 < \pi$ so $x/2$ lies in the second quadrant and $\cos(x/2) < 0$

$\implies$ f is increasing on the interval $(-\pi, \pi)$, and f is decreasing on the intervals $(-2\pi, -\pi)$ and $(\pi, 2\pi)$.

(e) Maxima/minima:

The first derivative test states we have a local maximum if the sign of f' changes from positive to negative, and a local minimum if the sign of f' changes from negative to positive.

Therefore there is a local maximum at $x = \pi$ and local minimum at $x = -\pi$ by the results in part (d).

(f) Intervals of concavity: Concavity can change at interior points in $[-2\pi, 2\pi]$ where f'' is undefined or equal to 0 are only *candidates* for the inflection points.

$$f''(x) = \frac{-3}{2}\sin\left(\frac{x}{2}\right) \cdot \frac{1}{2} = \frac{-3}{4}\sin\left(\frac{x}{2}\right)$$

$\frac{-3}{4}\sin\left(\frac{x}{2}\right) = 0$ when $\sin\left(\frac{x}{2}\right) = 0$ which in the domain of $-2\pi < x < 2\pi$ occurs at $x/2 = n\pi$ with n an integer. $-2\pi < x < 2\pi$ and $x = n2\pi$ with n an integer $\implies$ $x = 0$

We have a candidate for an inflection point at $x = 0$.
If $-2\pi < x < 0$, it follows that $-\pi < x/2 < 0$ so $x/2$ lies in the third and fourth quadrant and $\sin(x/2) < 0$.
If $0 < x < 2\pi$, it follows that $0 < x/2 < \pi$ so $x/2$ lies in the first and second quadrant and $\sin(x/2) > 0$.

So f is concave up on the interval $(-2\pi, 0)$, and f is concave down on the interval $(0, 2\pi)$.

(g) Inflection points: Inflection points occur where the concavity of a function changes. Those interior points in $[-2\pi, 2\pi]$ where f'' is undefined or equal to 0 are only *candidates* for the inflection points. We found in part f that we had a candidate for an inflection point at $x = 0$. We verified in part f that concavity changes at $x = 0$.

Since $f(0) = 3\sin\left(\frac{0}{2}\right) = 0$, the inflection point is $(0, 0)$

(h) Sketch the graph: